

PALM: A quasi-Poisson regression approach for large-scale microbiome association studies and meta-analyses

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Abstract

Identifying key microbial features associated with clinical outcomes, host factors, and other covariates is central to advancing microbiome research. Recent microbiome association studies have scaled up significantly, incorporating a greater number of microbial features, covariates, and datasets from diverse populations and cohorts. The unique characteristics of microbiome data, such as compositionality, sparsity, and overdispersion, pose significant challenges for association analysis, particularly in large-scale and meta-analytic contexts, often leading to low replicability of findings. To address these challenges, we introduce PALM, a quasi-Poisson regression framework designed for fast and reliable microbiome association discovery in large-scale studies and meta-analyses. Extensive realistic simulations demonstrate the advantages of PALM in false discovery rate control, statistical power, computational efficiency, and cross-study signal harmonization. The utility of PALM is illustrated through three real data applications: a meta-analysis of microbiome studies for colorectal cancer, a meta-analysis linking the microbiome to hundreds of metabolites, and a study associating the microbiome with millions of genetic variations in the host genome.

Significance Statement

High-throughput sequencing technology empowers researchers to profile microbiomes, yet the resulting data are highly complex, capturing only the noisy relative abundances of various microbes. Uncovering key microbial associations with clinical outcomes, host factors, and other covariates is important for advancing precision medicine, understanding disease mechanisms, and identifying potential diagnostic and therapeutic targets in microbiome research. However, current methods often suffer from low statistical power, false-positive signals, and computational inefficiencies—challenges that are magnified in large-scale studies and meta-analyses. Our work introduces PALM, a novel method that provides accurate, scalable, and reliable solutions for microbiome association discovery, paving the way for more robust insights into the microbial underpinnings of health and disease.

Introduction

Significant progress has been made in understanding the role of the human microbiome in health and disease^{1,2}. Feature-level association analyses, commonly known as differential abundance analyses, are widely used to identify microbial features associated with clinical outcomes, host factors, and other relevant covariates. Recent advances have expanded microbial feature measurements beyond 16S rRNA gene amplicon profiling to multi-omics approaches, including metagenomics and metatranscriptomics^{3,4}. Additionally, microbiome studies are increasingly associating microbial features with an ever-growing number of covariates.^{5,6} For example, microbiome genome-wide association studies (mbGWAS) link millions of host genetic variants to individual microbial features. Furthermore, meta-analyses of microbiome association studies are becoming more common, aiming to enhance discovery power and identify microbial signatures that are generalizable across cohorts and populations⁷⁻¹⁰. The substantial growth in study scale offers an unprecedented opportunity to uncover novel microbial signatures, deepen our understanding of microbiome functions, and translate these insights into actionable public health interventions. However, low replicability remains a significant barrier to achieving these goals. For instance, while existing mbGWAS have identified numerous genetic loci associated with microbial features, only a handful of the associations were shared across studies¹¹.

The issue of low replication is partly attributable to the analytical challenges posed by the unique characteristics of microbiome data. Microbiome data consists of sequence read counts assigned to microbial features, which are systematically distorted from their true absolute abundances (AA) within a unit volume of an ecosystem. Because sequencing depth is uncorrelated with the actual microbial load (the sum of AA across all features) in a sample, these counts only provide meaningful information about relative abundances (RA)^{12,13}. These RA count data are inherently noisy. Measurement biases are introduced during experimental workflows (e.g., DNA extraction, primer binding, and amplification), leading to differential efficiencies in detecting features¹⁴. The majority of counts are concentrated in a few abundant features, with the remainder distributed among numerous low-abundance

features. This results in a high proportion of zero counts, with the likelihood of observing zeros heavily influenced by a sample’s sequencing depth. Zero counts can also arise from true feature absence, which is difficult to distinguish from zeros due to insufficient sampling. The RA counts also exhibit substantial variation across samples, particularly when generated from different sequencing batches, study designs, or populations. These unique characteristics make microbiome data difficult to model.

Despite extensive research on microbiome association analysis, existing methods continue to face major issues, including high false positive rates (FDR) and inaccurate or unstable association effect inference. In meta-analyses, the estimated association effects of individual studies often show significant heterogeneity, meaning variation in effect sizes across studies exceeds what would be expected by chance^{9,15}. The premise of generalizability assumes that true effects are universal across studies, even though microbial community structures often vary substantially due to differences in technical batches, cohort populations, study designs, and other factors¹⁶. These unwanted variations, unrelated to the covariate of interest, are often referred to as batch effects. If not properly addressed in the analysis, batch effects can distort the associations of interest. Notably, inaccurate and/or unstable effect inference could make the effects across studies appear heterogeneous, even when the true effects are homogeneous. Addressing these issues would improve the harmonization of association signals across studies and minimize false alarms of inconsistent effects, ultimately leading to more powerful, robust, and generalizable findings.

These issues may stem from several sources: (1) Data normalization: Many methods rely on normalization techniques such as zero imputations, transformations (e.g., centered log-ratio), and batch effect corrections. However, normalization may fail to fully remove unwanted effects and may even distort association signals of interest^{17,18}. (2) Compositionality: Many methods focus on detecting RA-level associations. Since a change in the AA of a single feature shifts the RA of all features, compositional effects can obscure true associations. Correcting for compositional effects and recovering AA-level associations can improve the accuracy of identifying true signals and better control FDR^{19,20}. (3) Statistical model

assumptions and inference approaches: Misspecifying the distribution of microbiome data can lead to biased effect estimation and poor false discovery rate (FDR) control. Even with correctly specified models, FDR inflation and inaccurate effect estimates may still occur, particularly when analyzing highly sparse and variable microbiome data with a moderate sample size.

In this article, we introduce a semi-parametric quasi-Poisson method for Association analysis of Large-scale Microbiome studies and meta-analysis (PALM). PALM provides fast and reliable estimates of AA-level association effects and their variance. In meta-analyses, PALM integrates the AA-level summary statistics across studies, improving the harmonization of association effects and enhancing the generalizability of findings across studies. We evaluate the performance of PALM through extensive realistic simulation studies and illustrate its utility in three real data applications: identifying microbial features associated with colorectal cancer, hundreds of metabolites, and millions of host genetic variations. These studies highlight the superior performance of PALM in FDR control, discovery power, computational efficiency, and signal harmonization across studies.

Method

We first elucidate the relationship between RA and AA associations, laying the foundation for PALM’s approach to recovering AA associations. We then introduce the PALM framework in the context of meta-analysis. A schematic flowchart illustrating the methodology is provided in *SI Appendix*, Fig. S1.

Bridging RA associations to AA associations

Consider a study with n independent samples. Let K denote the number of features in the microbiome data. For each sample $i = 1, \dots, n$, let X_i denote the covariate of interest, N_i denote the sequencing depth of the microbiome, Y_{ik} denote the number of sequencing reads assigned to feature k (i.e., RA count data), W_{ik} denote the unobserved AA of feature k in a unit volume of the ecosystem, and O_i denote the microbial load, defined as $O_i = \sum_{k=1}^K W_{ik}$.

134 It is natural to model the mean of W_{ik} as

$$135 \quad E(W_{ik}) = \exp(\beta_{0k} + \beta_k X_i),$$

136 where β_{0k} is the intercept and β_k is the covariate effect on AA, referred to as the AA-level
 137 association effect. To reflect the process by which sequencing experiments generate RA
 138 count data, the conditional mean of Y_{ik} given O_i can be expressed as

$$139 \quad E(Y_{ik} | O_i) = N_i b_k \frac{E(W_{ik} | O_i)}{O_i} \approx N_i b_k \frac{E(W_{ik})}{O_i},$$

140 where b_k is a feature-level multiplicative factor accounting for measurement biases, such
 141 as differences in DNA extraction efficiency, primer binding, and amplification efficiency
 142 in measuring various features¹⁴. The approximation $E(W_{ik}) \approx E(W_{ik} | O_i)$ holds for
 143 high-dimensional microbiome data with a large number of features, provided that no single
 144 feature overwhelmingly dominates the community and dependencies among features are
 145 weak (*SI Appendix*). These conditions are generally met in typical microbiome datasets.

146 Since O_i is the summation of W_{ik} , it would also be associated with X_i if any W_{ik} is associated
 147 with X_i . Therefore, we can express O_i as $O_i = \exp(\bar{O}_i + \beta_O X_i)$, where β_O is the covariate
 148 effect on O_i and $\bar{O}_i$ is the residual on the log scale. The conditional mean formula can then
 149 be rewritten as

$$150 \quad E(Y_{ik} | O_i) = N_i \exp(\beta_{0k}^* + \beta_k^* X_i - \bar{O}_i), \quad (1)$$

151 where $\beta_{0k}^* = \beta_{0k} + \log(b_k)$ and $\beta_k^* = \beta_k - \beta_O$.

152 The β_k^* is the covariate effect on RA, referred to as the RA-level association effect. Note
 153 that the RA-level association effects for all features $k = 1, \dots, K$ are simply their AA-level
 154 association effects shifted by β_O . Since O_i is unobserved, we work with the marginal mean
 155 model for RA count expressed as

$$157 \quad E(Y_{ik}) = N_i \exp(\beta_{0k}^\dagger + \beta_k^* X_i), \quad (2)$$

158 where $\beta_{0k}^\dagger = \beta_{0k}^* + \log E(e^{-\bar{O}_i})$.

160 Clearly, the unidentifiable factors (i.e., b_k , $E(e^{-\bar{O}_i})$) are absorbed into the intercept, while
 161 the RA-level association effect β_k^* remains unchanged from the conditional model (1).

Generating summary statistics for individual studies

To estimate β_k^* , we propose using the quasi-Poisson regression with the marginal mean specified in (2) and variance $Var(Y_{ik}) = \phi E(Y_{ik})$, where ϕ is the dispersion parameter. This semi-parametric approach accounts for the overdispersion of RA count data but does not rely on specific distributional assumptions. Let $\boldsymbol{\theta}_k = (\beta_{0k}^\dagger, \beta_k^*)$, $\mathbf{X}_i = (1, X_i)$, and $\mu_{ik} = N_i \exp(\mathbf{X}_i^\top \boldsymbol{\theta}_k)$. We employ the Firth bias-corrected²¹ Poisson quasi-score function

$$\mathbf{S}(\boldsymbol{\theta}_k) = \sum_{i=1}^n \left(Y_{ik} - \mu_{ik} + \frac{1}{2} h_{ik} \right) \mathbf{X}_i.$$

The $\mathbf{X}_i$ can be expanded to include potential confounders. The h_{ik} is the i th diagonal element of the ‘hat’ matrix $\mathbf{H} = \mathbf{W}^{1/2} \mathbf{X} (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W}^{1/2}$, where the $\mathbf{W}$ is a diagonal matrix with diagonal elements $(\mu_{1k}, \dots, \mu_{nk})$ and $\mathbf{X}$ is the design matrix. The Firth correction effectively reduces bias in parameter estimates, particularly in cases of small sample sizes and features with a high proportion of zero counts²¹.

We can estimate β_k^* by solving $\mathbf{S}(\boldsymbol{\theta}_k) = \mathbf{0}$. However, fitting a separate model for each feature-covariate pair becomes computationally slow when the number of features and covariates is large. To address this, we propose estimating β_k^* using the score statistic. Let $\tilde{\boldsymbol{\theta}}_k$ denote the estimate of $\boldsymbol{\theta}_k$ under the null model, which is independent of the covariate of interest. Partition the quasi-score function $\mathbf{S}(\boldsymbol{\theta}_k)$ into $(S_1, \mathbf{S}_0)$, corresponding to the parameter of interest β_k^* and the nuisance parameters, respectively. Similarly, partition the quasi-information matrix $-\frac{\partial \mathbf{S}(\boldsymbol{\theta}_k)}{\partial \boldsymbol{\theta}_k} = \sum_{i=1}^n \mu_{ik} \mathbf{X}_i \mathbf{X}_i^\top$ as $\begin{bmatrix} \mathcal{I}_{11} & \mathcal{I}_{10} \\ \mathcal{I}_{01} & \mathcal{I}_{00} \end{bmatrix}$. The RA-level association effect β_k^* is estimated using the score statistic

$$\hat{\beta}_k^* = \mathcal{I}_1^{-1}(\tilde{\boldsymbol{\theta}}_k) S_1(\tilde{\boldsymbol{\theta}}_k),$$

where $\mathcal{I}_1 = \mathcal{I}_{11} - \mathcal{I}_{10} \mathcal{I}_{00}^{-1} \mathcal{I}_{01}$. This estimate is derived from the Taylor series expansion of the quasi-score function. Score-statistic-based association analysis is widely used in modern GWAS due to its greater statistical accuracy and numerical stability compared to Wald-statistic-based analysis^{22,23}.

187 The variance of $\widehat{\beta}_k^*$ can be robustly estimated using the sandwich formula²⁴

$$188 \quad V_k^* = \mathcal{I}_1^{-1}(\tilde{\boldsymbol{\theta}}_k) \left(\sum_{i=1}^n U_i(\tilde{\boldsymbol{\theta}}_k) U_i^\top(\tilde{\boldsymbol{\theta}}_k) \right) \mathcal{I}_1^{-1}(\tilde{\boldsymbol{\theta}}_k),$$

189 where $U_i = S_{1,i} - \mathcal{I}_{10} \mathcal{I}_{00}^{-1} \mathbf{S}_{0,i}$, $S_{1,i}$ and $\mathbf{S}_{0,i}$ are the i th summand of S_1 and $\mathbf{S}_0$, respectively.

190 For studies with correlated samples (e.g., longitudinal data, repeated measurements, or
191 family-based data), the PALM summary statistics can be obtained in a similar manner,
192 with adjusted variance estimates to account for correlation among samples within a cluster
193 (*SI Appendix*).

194 Recovering AA-level summary statistics

195 We are interested in detecting AA-level associations ($H_0 : \beta_k = 0$). Given the relationship
196 between AA- and RA-level association effects, $\beta_k^* = \beta_k - \beta_O$, it follows that β_O represents
197 the common compositional effect shared by all features ($k = 1, \dots, K$). Without additional
198 assumptions, β_O is not estimable, and AA-level effects are unidentifiable from RA-level
199 effects. Here, we leverage the high dimensionality of microbial features and assume that the
200 majority of features have no AA-level effects (i.e., $\beta_k = 0$ for most k). Under this sparse
201 AA signal assumption, we propose estimating $-\beta_O$ using $\text{median}\{\widehat{\beta}_k^*\}_k$ and obtaining the
202 estimate of the AA-level effect as

$$203 \quad \widehat{\beta}_k = \widehat{\beta}_k^* - \text{median}\{\widehat{\beta}_k^*\}_k.$$

204 Alternative statistics, such as the inverse-variance weighted mean of $\widehat{\beta}_k^*$'s, can also be used
205 to estimate the compositional effect. A key advantage of using the median is its robustness
206 to extreme values and outliers.

207 The variance of $\widehat{\beta}_k$ can be calibrated from V_k^* by taking into account the variability of the
208 median as (*SI Appendix*)

$$209 \quad V_k \approx V_k^* + \frac{K}{4 \left(\sum_{k=1}^K \frac{1}{\sqrt{2\pi V_k^*}} \right)^2}.$$

210 The $\widehat{\beta}_k$ and V_k can be conveniently shared as AA-level summary statistics across studies
211 and subsequently combined to perform meta-analysis for association testing.

Meta-analysis

Suppose we want to meta-analyze L microbiome association studies. For study $\ell = 1, \dots, L$, we obtain AA-level association effect estimate $\hat{\beta}_{k\ell}$ and its variance $V_{k\ell}$ for each feature k . A feature may be unobserved or excluded from a given study due to its low prevalence in that study. We define $\mathcal{L}_k$ as the set of indices for studies where feature k is observed. Under the fixed-effect model, the inverse-variance weighted estimate of the overall effect is

$$\hat{\beta}_{k,meta} = \frac{\sum_{\ell \in \mathcal{L}_k} \hat{\beta}_{k\ell} / V_{k\ell}}{\sum_{\ell \in \mathcal{L}_k} 1 / V_{k\ell}}.$$

The variance of $\hat{\beta}_{k,meta}$ is estimated as

$$V_{k,meta} = \frac{1}{\sum_{\ell \in \mathcal{L}_k} 1 / V_{k\ell}}.$$

Finally, we construct the test statistic $Q_{k,meta} = \hat{\beta}_{k,meta}^2 / V_{k,meta}$ and calculate the p -value using the chi-square distribution with one degree of freedom.

In summary, the modeling and estimation procedure of PALM offers a unique combination of advantages: (i) No normalization requirements: PALM directly models RA data, avoiding biases and instability introduced by data normalization. (ii) No distributional assumptions: PALM employs a semi-parametric approach that accommodates overdispersion in RA count data without relying on strict distributional assumptions. This results in more reliable inference compared to parametric models. (iii) Compositional effect correction: PALM connects association effects in the RA model to those in the latent AA model. By leveraging this relationship and the high dimensionality of microbial features, PALM accurately corrects for compositional effects and recovers AA-level summary statistics. (iv) Fast and stable inference: PALM constructs summary statistics using score statistics, which improves numerical stability and accuracy for highly sparse and variable microbiome data compared to Wald statistics. Importantly, these score statistics are computed by fitting a single null model independent of the covariate of interest, making PALM computationally efficient for association studies with large numbers of features and covariates. (v) Flexibility: PALM supports diverse study designs, such as cross-sectional and longitudinal studies, and allows for adjustment of confounders.

Results

Simulation studies

We evaluated the performance of PALM and compared it with several leading methods in the meta-analysis of five studies testing for differential AA between two comparison groups. Each method generated summary statistics, which were combined using a fixed-effect meta-analysis, as the simulated association effects were homogeneous across studies. The compared methods included ANCOM-BC2¹⁹ and LinDA²⁰, designed to detect differential AA. For ANCOM-BC2, we used the version that applies a feature-level sensitivity score filter to reduce potential false positives¹⁹. Therefore, only studies where a feature passed the sensitivity score contributed to the ANCOM-BC2 meta-analysis. Additionally, we included MaAsLin2⁵ and LM-CLR²⁵, which are widely used for detecting differential RA. LM-CLR is either a linear model for studies with independent samples or a linear mixed model for studies with correlated samples, and it uses the centered log-ratio (CLR)-transformed feature variable as the response. The details on the implementation of the compared methods are provided in *SI Appendix*.

We simulated microbiome data for five studies using five real metagenomics datasets as templates⁸. Real data from five studies showed systematic differences in microbial community structures (*SI Appendix* Fig. S2a). To generate realistic simulated data, we used the recently introduced microbiome data simulator MIDAsim²⁶. We considered $K = 401$ species with a prevalence of at least 20% across the five studies in the real data and $K = 92$ genera derived by collapsing the species-level data. To learn the structure of species (or genus) counts in each study, we used the `MIDASim.setup` function in nonparametric mode. The function returned the mean proportion vector (μ) and additional parameters (e.g., correlation matrices) that define the structure of the template data. We simulated two equally sized groups of samples ($X = 1$ and $X = 0$) by introducing differential AA signals into the mean proportion vector between the groups. Specifically, we randomly selected a subset of features as “active” features. Half of these were chosen randomly from relatively abundant features, defined as having an average proportion of at least 10^{-3} in the original data, while the other half were randomly selected from the remaining features.

267 The mean proportion vectors for the two groups were initialized as $\mu_1 = \mu_0 = \mu$. For
 268 active features with positive effects, we increased their proportions in μ_1 , and for active
 269 features with negative effects, we increased their proportions in μ_0 . The fold increase for
 270 each active feature was randomly sampled from $\text{Uniform}(1, \Delta)$. After adjusting for all
 271 active features, μ_1 and μ_0 were normalized so that their elements summed to 1, ensuring
 272 they represented valid proportions. Note that the proportions of all features were shifted
 273 after this step. However, the active features were considered true differential AA features,
 274 driven by changes in the underlying AA between the two groups, which in turn caused
 275 the shift in RA for all features. Finally, we consolidated all parameters, including μ_1 and
 276 μ_0 , the specified sequencing depths, and the other parameters learned from the template
 277 data, using the `MIDASim.modify` function. We then generated samples for the two groups
 278 using the `MIDASim` function. This procedure was repeated for each study, resulting in
 279 realistic simulated datasets closely resembling the original data (*SI Appendix*, Fig. S3).
 280 The randomly selected active features and their associated fold changes during the spike-in
 281 procedure were consistent across studies, representing homogeneous association effects.

282 We considered a broad range of scenarios in our simulations: (1) Two settings for feature
 283 size: species-level ($K = 401$) or genus-level ($K = 92$) data; (2) Two settings for sample
 284 sizes over the five studies: large sample size ($n = 100, 120, 140, 160, 180$) or small sample
 285 size ($n = 20, 40, 60, 80, 100$); The factor Δ , which controls the size of the fold change in
 286 proportion, was set to 5 for the large sample size setting and 10 for the small sample size
 287 setting. (3) Two effect direction settings: balanced positive/negative effects (active features
 288 had equal probabilities of having positive/negative effects) or dominant positive effects (all
 289 active features had positive effects). (4) Two sequencing depth unevenness settings: even
 290 (sequencing depths of all samples were randomly drawn from the pool of sequencing depths in
 291 the real data) or uneven (further doubled the sequencing depth for the samples in a randomly
 292 picked group); (5) A range of active feature proportions from 0.05 to 0.2. Combining all
 293 these factors resulted in 64 simulation settings, and we generated 100 simulated datasets for
 294 each setting.

295 We also conducted another set of simulations to meta-analyze studies with correlated samples.
 296 First, we simulated independent samples as described above. Next, we assigned every two
 297 samples Y_{i1} and Y_{i2} from the same group to form cluster i . To introduce correlation, we
 298 simulated another independent sample Y_i^* under the same group as Y_{i1} and Y_{i2} . This
 299 independent sample was then mixed with Y_{i1} and Y_{i2} to create the correlated samples
 300 $Y_{i1}^* = (Y_i^* + Y_{i1})/2$ and $Y_{i2}^* = (Y_i^* + Y_{i2})/2$ for cluster i .

301 Simulation results

302 We evaluated all methods in terms of FDR control and discovery power (Fig. 1). For
 303 all methods, we applied the Benjamini-Hochberg (BH) procedure²⁷ for multiple testing
 304 correction and reported empirical FDR and power at a target FDR of 0.05. PALM was the
 305 only method that successfully controlled FDR across all scenarios. Unbalanced positive/negative
 306 effects and uneven sequencing depth exacerbated FDR inflation in the compared methods.
 307 PALM consistently maintained high power, even when the other methods exhibited FDR
 308 inflation.

309 Evaluating effect size heterogeneity across studies is crucial in meta-analysis as it provides
 310 key insights into the consistency and generalizability of study findings. To this end, we
 311 reported the proportion of features with significant heterogeneity (Fig. 1), determined
 312 using Cochran’s Q test ($q\text{-val} < 0.1$). In our simulations, the true AA-level effects were
 313 homogeneous; therefore, accurate summary statistics should exhibit no heterogeneity. PALM
 314 summary statistics showed no heterogeneity, demonstrating its ability to harmonize truly
 315 homogeneous association signals in meta-analysis and produce reliable, generalizable findings.
 316 Other methods exhibited varying degrees of effect heterogeneity, with ANCOM-BC2 summary
 317 statistics showing the highest level in most scenarios.

318 We also compared PALM meta-analysis with pooled-data analyses by other methods (*SI*
 319 *Appendix*, Fig. S4). Specifically, the original individual-level data from all studies were
 320 pooled together, and study indicators were adjusted as confounders in the association
 321 analysis. For ANCOM-BC2, the pooled-data analysis achieved better FDR control compared

322 to its meta-analysis, but this came at the cost of reduced power. This trade-off occurred
323 because the ANCOM-BC2 sensitivity score filter removed both false positives and true
324 positives in the pooled-data analysis. For LinDA, LM-CLR, and MaAsLin2, the FDR
325 inflation in the pooled-data analysis was similar to those observed in their meta-analysis.

326 Moving beyond studies with independent samples, we next evaluated the performance of
327 methods in meta-analyzing studies with correlated samples (Fig. 2). PALM effectively
328 controlled the FDR, maintained high power, and exhibited no effect heterogeneity across
329 studies. While the sensitivity score filter in ANCOM-BC2 effectively reduced FDR, it also
330 lowered power. LinDA, LM-CLR, and MaAsLin2 exhibited FDR inflation and false effect
331 heterogeneity across studies.

332 **Computation time**

333 We quantified the computation time of different methods in meta-analysis across varying
334 sample sizes and feature sizes as the number of covariates increased from 10^2 to 10^5 (Fig. 3).
335 LinDA, LM-CLR, and PALM demonstrated the highest computational efficiency, making
336 them suitable for large-scale association scans. Increasing the number of features and
337 covariates led to longer computation times for all methods, but the impact was minimal
338 for LM-CLR and PALM. MaAsLin2 exhibited moderate computational efficiency, while
339 ANCOM-BC2 had the highest computational demand, making both methods impractical
340 for association scans with more than 10^5 covariates.

341 **Meta-analysis of microbiome association studies for colorectal cancer**

342 We applied PALM to three real data applications. In the first application, we meta-analyzed
343 five metagenomics studies^{8,28–31} to identify gut microbial features associated with colorectal
344 cancer (CRC). These five studies were conducted on three continents and differed in sampling
345 procedures, sample storage, and DNA extraction protocols. The combined sample size
346 across studies was 574, comprising 284 CRC cases and 290 controls (*SI Appendix*, Table
347 S1). All raw sequencing data were reprocessed using a unified bioinformatics pipeline for

348 taxonomic profiling, as previously described⁸. The microbial community structures across
349 studies showed significant differences (PERMANOVA test p -value $< 10^{-5}$, *SI Appendix*, Fig.
350 S2a). After filtering out microbial species with less than 20% prevalence across all samples,
351 401 species were retained for analysis.

352 ANCOM-BC2, LinDA, LM-CLR, MaAsLin2, and PALM identified 117, 120, 123, 126, and
353 84 features, respectively (Fig. 4a, *SI Appendix*, Table S2). A core set of 44 species was
354 consistently identified by all methods, including well-known microbial species associated with
355 CRC, such as *Fusobacterium nucleatum*, *Bacteroides fragilis*, and *Parvimonas micra*^{32,33}.
356 Notably, the majority of features identified by PALM (74%) were also selected by at least
357 three other methods. In contrast, ANCOM-BC2 identified 38 features that were not supported
358 by other methods. Most of these features (20) had extremely low abundance, with an average
359 proportion less than 5×10^{-4} . Furthermore, many of these features (15) failed to pass the
360 sensitivity score filter when ANCOM-BC2 was applied to pooled-data analysis, suggesting
361 they may be potential false signals.

362 PALM exclusively identified six features, including *Faecalibacterium prausnitzii*, species
363 under *Phascolarctobacterium*, and unknown members of *Ruminococcaceae*, *Clostridiales*,
364 and *Clostridium*. *Faecalibacterium prausnitzii* and certain members of *Ruminococcaceae*
365 are well-recognized as protective microbes against CRC^{31,34}. These microbes contribute to
366 gut health by producing short-chain fatty acids and promoting anti-inflammatory pathways
367 that help maintain gut barrier integrity and modulate immune responses^{35,36}. Recent
368 research has also suggested that *Faecalibacterium prausnitzii* may enhance the effectiveness
369 of immunotherapy for CRC while mitigating its adverse effects³⁷. For these taxa, PALM
370 effect estimates are consistently negative across studies, whereas effect estimates from other
371 methods showed both negative and positive associations (Fig. 4c, lower panel).

372 In general, the summary statistics from different methods exhibited varying levels of cross-study
373 association effect heterogeneity. PALM summary statistics showed no effect heterogeneity
374 across all features, whereas other methods displayed heterogeneity in some features (Cochran's

375 Q test q -value < 0.1), with the strongest heterogeneity observed in ANCOM-BC2 summary
376 statistics (Fig. 4b, c). While it is impossible to know whether true effect sizes are heterogeneous
377 in real data, our simulation results suggested that high observed heterogeneity may arise
378 from inaccurate effect inference, even when the true effect sizes are homogeneous.

379 **Meta-analysis of microbiome studies linking to the metabolome**

380 In the second application, we conducted a meta-analysis of eight microbiome-metabolome
381 association studies^{15,38–44} to identify gut microbial features associated with gut metabolites.
382 The total sample size across all studies was 2,127. The studies differed in microbiome
383 sequencing platforms, metabolomic measurement methods, cohort demographics, and study
384 designs (*SI Appendix*, Table S3). All raw sequencing data were reprocessed and metabolite
385 identifiers were consolidated for uniformity, as described elsewhere⁴⁵. The microbial community
386 structures across studies showed significant differences (PERMANOVA test p -value $< 10^{-5}$,
387 *SI Appendix*, Fig. S2b). Most metabolites were quantified in only a subset of studies.
388 To ensure better data quality, we focused on metabolites and genera with at least 10%
389 prevalence within individual studies and excluded metabolites present in only one study.
390 The final datasets for analysis contained 450 metabolites and 101 genera, yielding 43,722
391 metabolite-genus pairs for the association scan. Summary statistics for individual studies
392 were generated by treating each metabolite as a covariate of interest and adjusting for disease
393 status as a confounder.

394 LinDA, LM-CLR, and MaAsLin2 identified significantly more features than PALM across
395 metabolites (Fig. 5a, *SI Appendix*, Table S4). While ANCOM-BC2 identified a comparable
396 number of features to PALM (Fig. 5a), the overlap between features identified by the
397 two methods was low (*SI Appendix*, Fig. S5). In addition, we evaluated the number
398 of associated metabolites across all features and characterized the top features associated
399 with a large number of metabolites in the results of each method (*SI Appendix*, Fig. S6).
400 Compared to other methods, the top features identified by PALM tend to be more abundant
401 across individuals and are considered part of the human core microbiota^{46,47}. We also found

that this trend was driven by positive associations rather than negative associations (*SI Appendix*, Fig. S6b,c). Indeed, many of these features, such as *Phocaeicola*, *Bacteroides*, *Anaerostipes*, *Blautia*, *Faecalibacterium*, and *Roseburia*, are well-known producers of various metabolites, particularly short-chain fatty acids and other bioactive compounds essential for host health^{48–50}.

We further investigated cross-study effect heterogeneity in the summary statistics generated by different methods. For almost all metabolites, LinDA, LM-CLR, and MaAsLin2 identified a larger proportion of features with heterogeneous effects compared to PALM (Fig. 5b). As an example, we highlighted the association results for butyric acid (Fig. 5c; additional results for cholic acid are shown in *SI Appendix*, Fig. S7). Many features associated with butyrate exhibited significant effect heterogeneity in the summary statistics produced by LinDA, LM-CLR, and MaAsLin2. The lower panel of Fig. 5c presents results for three well-known taxa contributing to butyrate production. These taxa were identified by LinDA, LM-CLR, MaAsLin2, and PALM but were missed by ANCOM-BC2. Their PALM effect estimates were consistently positive across studies, but the ANCOM-BC2 effect estimates had inconsistent directions.

Analysis of microbiome and host genetics associations

In the third application, we conducted a gut microbiome and host genetics association study in a single infant cohort. The cohort consisted of 502 healthy term infants, with their gut microbiomes measured once during their first year of life. In our analysis, the microbial features included 109 amplicon sequence variants (ASVs) with a prevalence greater than 5%, and the covariates comprised 6,174,920 SNPs with a minor allele frequency (MAF) > 5%, resulting in about 6.7×10^8 ASV-SNP pairs for the association scan. More details on the data are available in *SI Appendix*. We applied computationally feasible methods LinDA, LM-CLR, and PALM to perform the mbGWAS. Several existing mbGWAS studies have also used LM-CLR in their analyses^{25,51}. In all methods, we adjusted for age at enrollment, sex, the top five ancestry principal components, delivery mode, antibiotic exposure, and birth

429 season as confounders.

430 Our analyses were completed within 20 hours using 48 parallel cores on a 2.30 GHz Intel
431 Xeon Gold 5118 processor. Using the genome-wide significance threshold ($p\text{-value} < 5 \times$
432 10^{-8}), LinDA identified 71 ASVs associated with at least one SNP, LM-CLR identified 50
433 ASVs, and PALM identified a single ASV (ASV2) associated with three SNPs (the top
434 SNP is rs7780824 with $p\text{-value} = 6.4 \times 10^{-9}$, Fig 6a). Many ASVs associated with a large
435 number of SNPs in the LinDA and LM-CLR results tend to have relatively low abundance
436 (*SI Appendix*, Fig. S8a). Both methods rely on the CLR transformation, which requires
437 adding a pseudo-count to handle zeros. To assess the impact of the pseudo-count, we re-ran
438 LinDA and LM-CLR, varying the pseudo-count from the default value of 0.5 to 0.01. While
439 both methods still identified many significant hits (*SI Appendix*, Fig. S8b), the majority
440 differed from those identified previously (Fig. 6b), suggesting that the findings in LinDA
441 and LM-CLR were unstable and may include false positives. The PALM-identified ASV2
442 belongs to genus *Escherichia-Shigella* and is highly prevalent in the cohort (92%), which
443 naturally enhances its statistical power for detection in a moderate-sized GWAS. A clear
444 trend of increased average relative abundance of ASV2 is observed in individuals with higher
445 alternative allele dosages for rs7780824 (Fig. 6c).

446 Discussion

447 The unique characteristics of microbiome data pose significant analytical challenges, which
448 contribute to the poor replicability of findings, particularly in large-scale association studies
449 and meta-analyses. We introduce PALM, a statistical framework designed for fast and
450 reliable association discovery. PALM directly models feature RA counts using quasi-Poisson
451 regression to infer AA-level associations without requiring data normalization or distributional
452 assumptions. By leveraging score statistics, PALM achieves computational efficiency and
453 reliable association effect inference. Furthermore, PALM harmonizes homogeneous association
454 signals across studies, even when microbial communities exhibit high variability due to
455 differences in technical batches, populations, study designs, and other factors.

456 We conducted comprehensive and realistic simulation studies to validate the performance of
457 PALM in the meta-analysis of microbiome association studies. Recent research highlights
458 that unrealistic simulations can compromise the fair benchmarking of microbiome differential
459 abundance methods⁵², underscoring the importance of assessing the similarity between
460 simulated and real data. In our simulations, we used five real microbiome datasets as
461 templates to generate realistic multi-cohort microbiome datasets. The simulated data
462 closely mirrored the structure of the real data, providing strong support for the reliability
463 of our method comparisons. The simulations showed that PALM was the only method
464 to consistently control FDR across diverse scenarios. It also demonstrated high power in
465 identifying true covariate-associated features at the AA level. In our simulations, where
466 the introduced association effects were truly homogeneous across studies, PALM was the
467 only method to provide homogeneous association effect estimates across all scenarios. The
468 performance of PALM in real data applications aligns with these simulation results. In
469 the meta-analyses, PALM summary statistics exhibited lower cross-study heterogeneity
470 compared to those of other methods. While PALM generally identified fewer microbial
471 features than other methods, its discoveries were more stable and biologically meaningful.

472 The design of PALM enables it to efficiently handle association analyses involving a large
473 number of features and covariates. Unlike many existing methods, PALM requires fitting a
474 null model only once, regardless of the number of covariates, making it highly computationally
475 scalable. This advantage was demonstrated in the mbGWAS application, where PALM
476 efficiently analyzed millions of genetic variants. Importantly, its computational efficiency
477 does not compromise statistical accuracy, establishing PALM as a powerful tool for large-scale
478 microbiome studies.

479 The current implementation of PALM is based on the quasi-Poisson model, which assumes
480 a linear relationship between the variance and the mean. However, this mean-variance
481 relationship may not hold for all microbial features. The PALM framework could be
482 extended to incorporate the quasi-gamma-Poisson model, allowing the variance to follow
483 a quadratic relationship with the mean. This article focuses on testing associations with

one covariate at a time. Extending PALM to support multi-covariate association analyses would broaden its utility and application scope. In summary, PALM addresses critical methodological gaps in microbiome association studies by integrating robust statistical modeling, computational scalability, and effective meta-analysis capabilities. By delivering reliable and generalizable results, PALM will accelerate discoveries with translational potential and advance microbiome research in health and disease.

490 **Data availability.** The datasets for the five CRC studies were obtained from [https://](https://github.com/zellerlab/crc_meta)
491 github.com/zellerlab/crc_meta. The datasets for the eight MTBL studies were obtained
492 from <https://github.com/borenstein-lab/microbiome-metabolome-curated-data>. The
493 datasets for the host genetics study are available upon request by contacting the co-authors
494 at Vanderbilt. PALM is implemented in the R package PALM publicly available at [https:](https://github.com/ZjpWei/PALM_package)
495 [//github.com/ZjpWei/PALM_package](https://github.com/ZjpWei/PALM_package). The site also contains a complete set of manuals and
496 instructions. The scripts generating reported results are deposited at [https://github.com/](https://github.com/ZjpWei/PALM)
497 [ZjpWei/PALM](https://github.com/ZjpWei/PALM).

498 **Acknowledgments**

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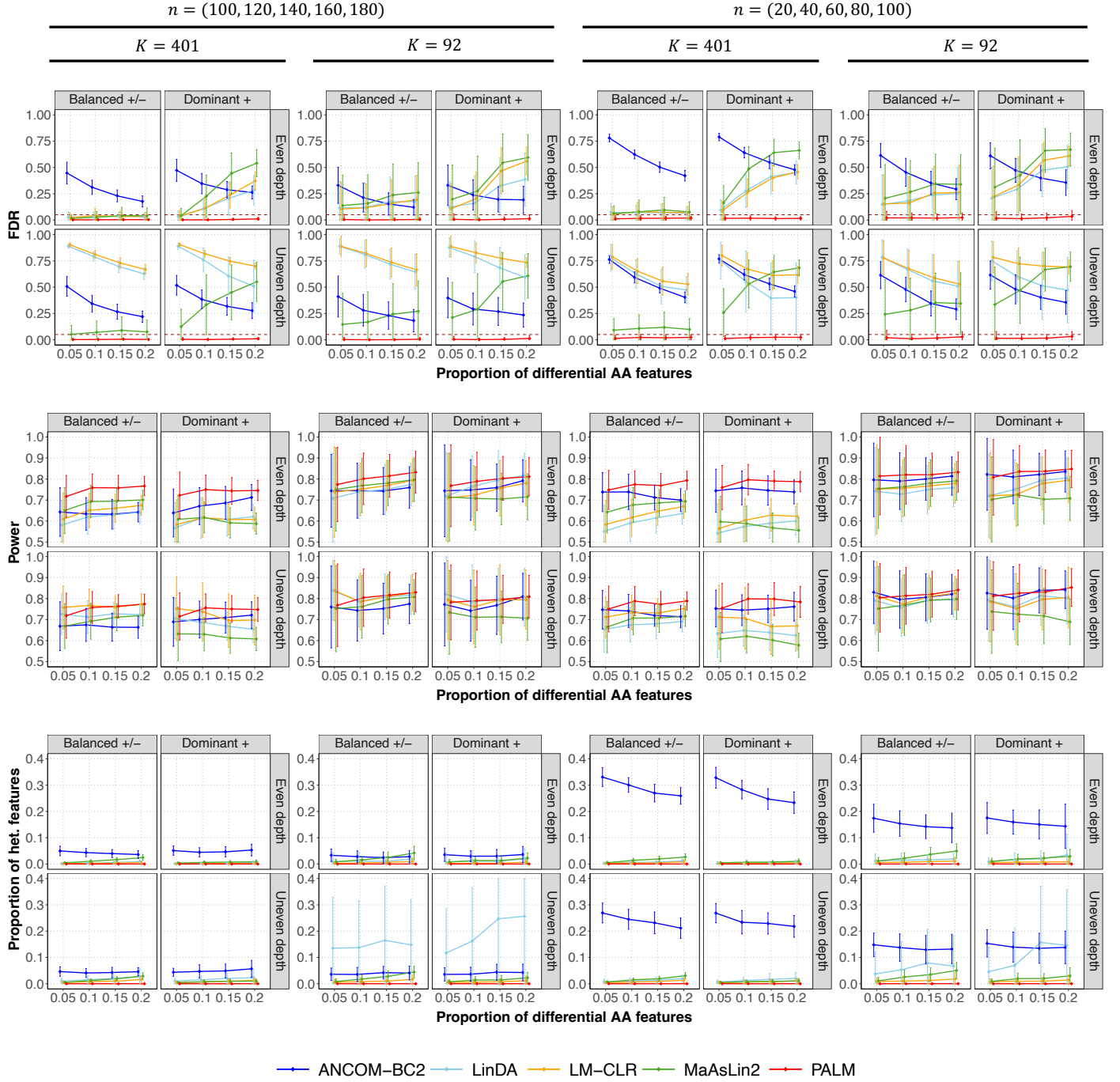


Fig. 1: Comparison of different methods in the meta-analysis of five simulated studies with independent samples. The figure is divided into three sections, each representing a benchmark metric (y-axis): empirical FDR (top; the horizontal line indicates the target FDR), power (middle), and the proportion of heterogeneous features (bottom). Heterogeneous features are defined as those with variation in effect sizes across studies exceeding what is expected by chance (Cochran's Q test q -value < 0.1). The x-axis represents the proportion of differential absolute abundance (AA) features. The top title shows the two scenarios of sample sizes n in five studies. The second title shows the two scenarios of feature size K . The column facet title represents two effect direction scenarios: Balanced +/-, where differential AA features have an equal probability of exhibiting positive or negative effects, and Dominant +, where all differential AA features exhibit positive effects. The row facet title represents two sequencing depth scenarios: one with even sequencing depth and one with uneven sequencing depth between the comparison groups. Each panel displays the mean estimated metrics with $\pm$ standard errors (indicated by error bars) based on 100 simulation replicates.

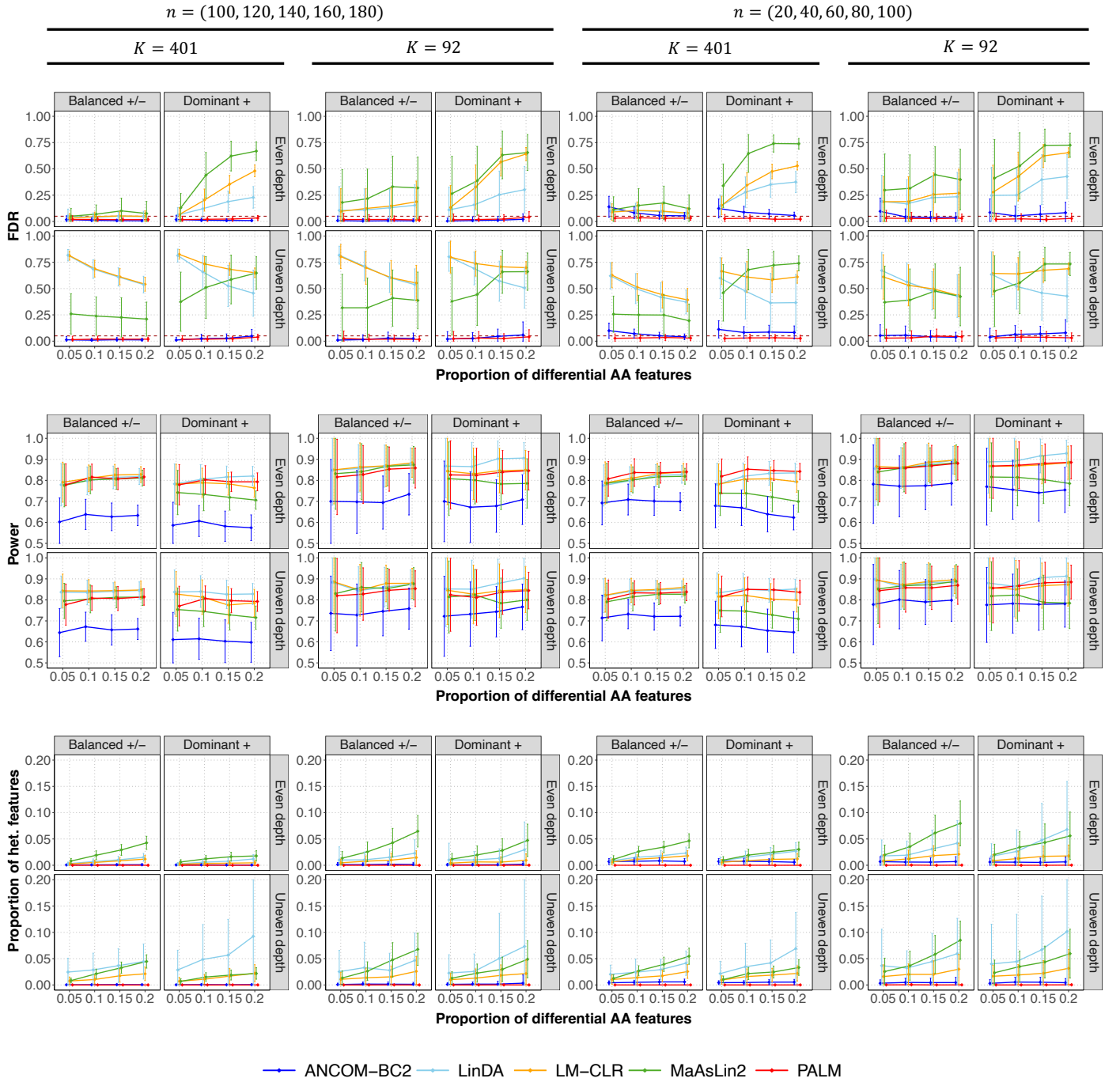


Fig. 2: Comparison of different methods in the meta-analysis of five simulated studies with correlated samples. The figure is divided into three sections, each representing a benchmark metric (y-axis): empirical FDR (top; the horizontal line indicates the target FDR), power (middle), and the proportion of heterogeneous features (bottom). Heterogeneous features are defined as those with variation in effect sizes across studies exceeding what is expected by chance (Cochran's Q test q -value < 0.1). The x-axis represents the proportion of differential absolute abundance (AA) features. The top title shows the two scenarios of sample sizes n in five studies. The second title shows the two scenarios of feature size K . The column facet title represents two effect direction scenarios: Balanced +/-, where differential AA features have an equal probability of exhibiting positive or negative effects, and Dominant +, where all differential AA features exhibit positive effects. The row facet title represents two sequencing depth scenarios: one with even sequencing depth and one with uneven sequencing depth between the comparison groups. Each panel displays the mean estimated metrics with $\pm$ standard errors (indicated by error bars) based on 100 simulation replicates.

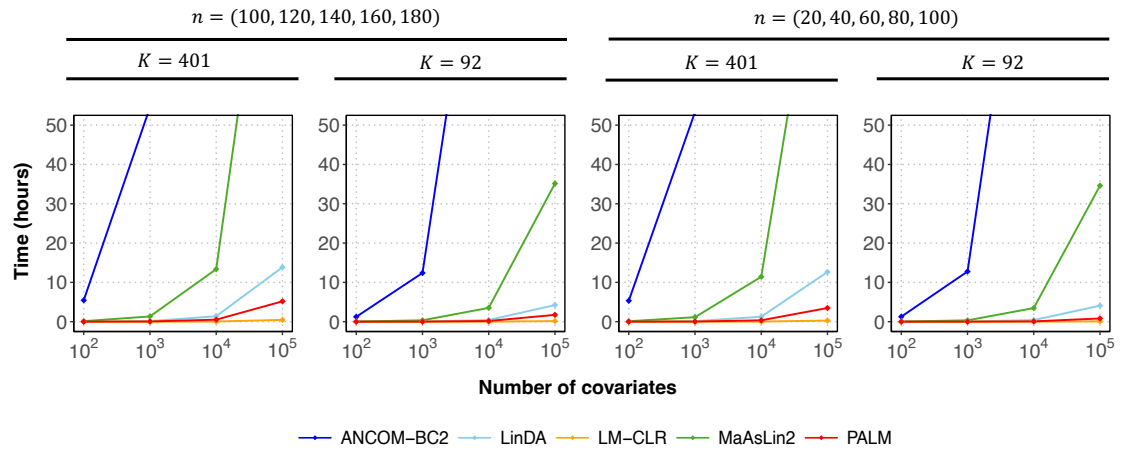


Fig. 3: Comparison of computation time of different methods in the meta-analysis of five simulated studies with independent samples. The top title shows the two scenarios of sample sizes n in five studies. The second title shows the two scenarios of feature size K . Each plot shows the computation time in hours of different methods as a function of the number of covariates. The association scan was performed for one covariate at a time, and no parallel computing was used for all methods. Time beyond 50 hours is truncated. Values are averages over 100 simulation replicates for each setting.

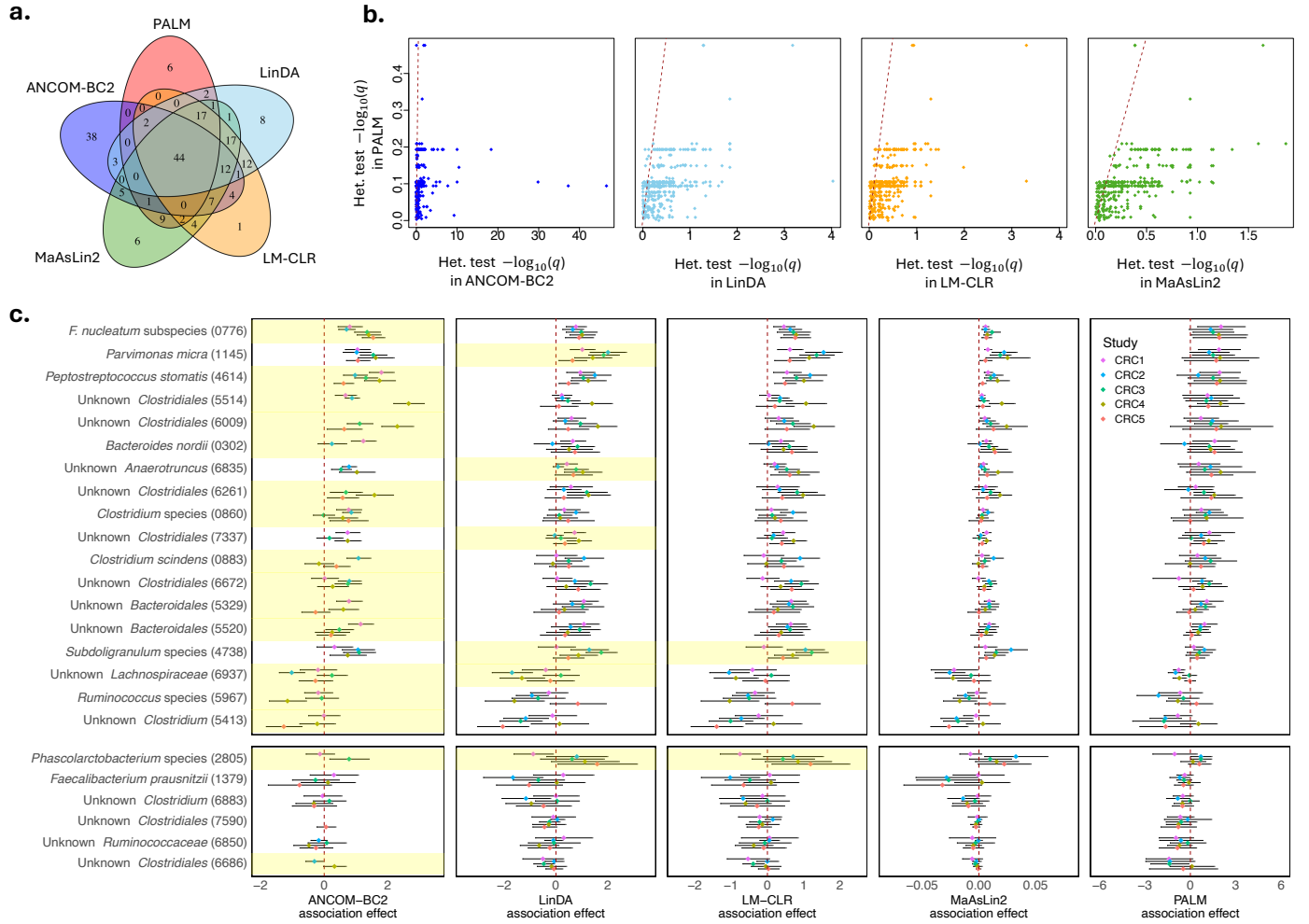


Fig. 4: Results of the meta-analysis of microbiome association studies for colorectal cancer (CRC). **a**, The Venn diagram shows the overlap of the CRC-associated features identified by different methods. **b**, Scatter plots of Cochran's Q heterogeneity test $-\log_{10}(q\text{-value})$ in the meta-analysis using summary statistics of PALM (y-axis) and those of ANCOM-BC2, LinDA, LM-CLR, and MaAsLin2 (x-axis) over all features (dots). **c**, Upper panel: x-axis represents a core set of CRC-associated features identified by all methods but exhibiting significant heterogeneity in effects across studies in at least one method. Each box shows the summary statistics of a method, including the association effect estimates across studies (color-coded dots) and their 95% confidence intervals (black lines). In the box for ANCOM-BC2, the effects of some studies were not shown, as they were removed by the sensitive score filter. The row for features with significant heterogeneous effects across studies is shaded yellow in each panel (Cochran's Q test $q\text{-value} < 0.1$); Lower panel: similar format as the upper panel but for the six features exclusively identified by PALM.

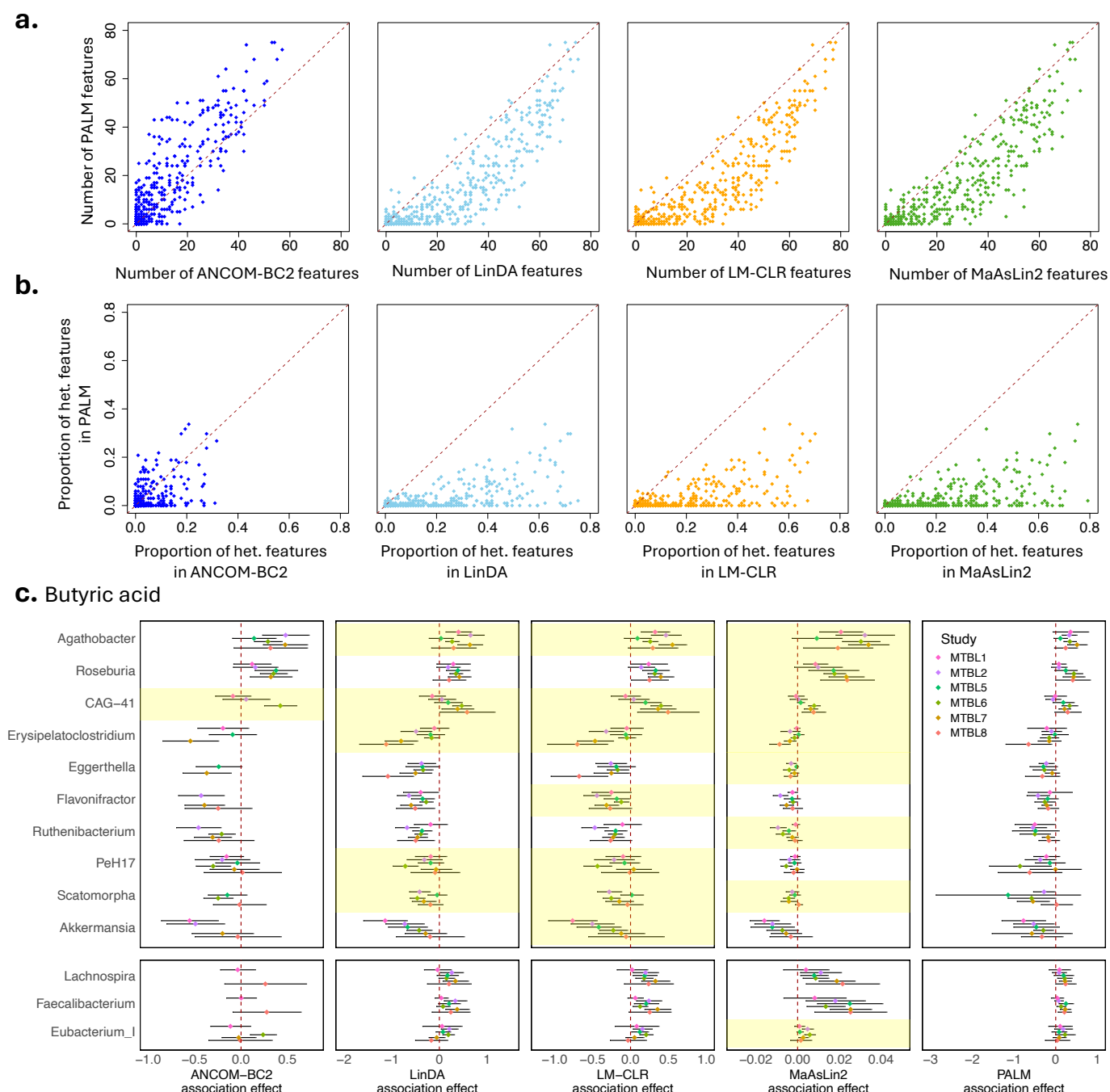


Fig. 5: Results of the meta-analysis of microbiome-metabolome (MTBL) association studies. **a**, Scatter plots of the number of features identified by PALM (y-axis) and by ANCOM-BC2, LinDA, LM-CLR, and MaAsLin2 (x-axis) over all metabolites (dots). **b**, Scatter plots of proportions of features with significant cross-study effect heterogeneity (Cochran's Q test q -value < 0.1) based on summary statistics of PALM (y-axis) and ANCOM-BC2, LinDA, LM-CLR, and MaAsLin2 (x-axis) over all metabolites (dots). **c**, Upper panel: x-axis represents a core set of butyric-acid-associated features identified by all methods but exhibiting significant heterogeneity in effects across studies in at least one method. Each box shows the summary statistics of a method, including the association effect estimates across studies (color-coded dots) and their 95% confidence intervals (black lines). In the box for ANCOM-BC2, the effects of some studies were not shown, as they were removed by the sensitive score filter. The row for features with significant heterogeneous effects across studies is shaded yellow in each panel (Cochran's Q test q -value < 0.1); Lower panel: similar format as the upper panel but for the three well-known butyric-acid-associated features not included in the upper panel.

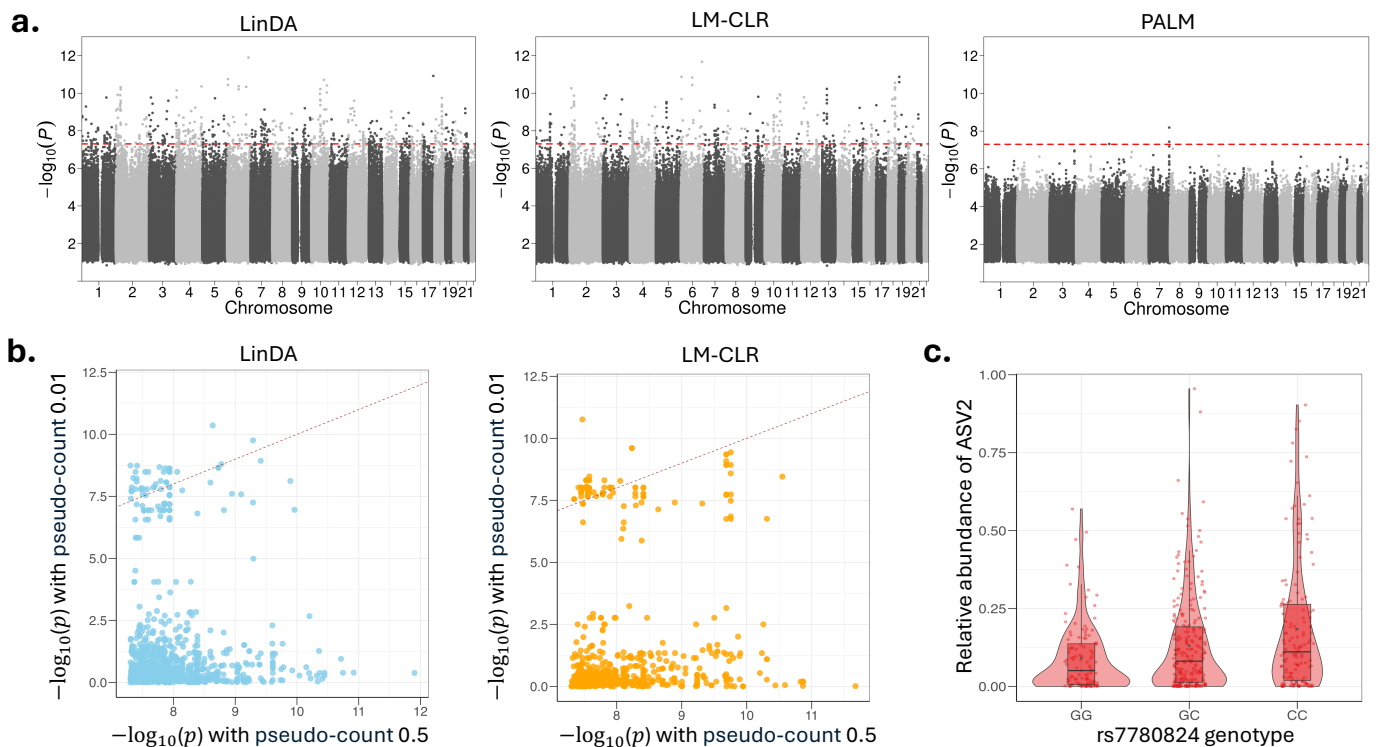


Fig. 6: Results of the microbiome and host genetics GWAS. **a**, Manhattan plot aggregating the top associations with the microbial feature. Each SNP was tested against each of the 109 ASVs and the Manhattan plot shows the lowest resulting p -value for each SNP. The horizontal dashed line marks the genome-wide significance (p -value = 5×10^{-8}). The top hit in PALM results involves feature ASV2 and SNP rs7780824. **b**, Comparison of p -values when using the pseudo-count of 0.5 (default) and 0.01 in LinDA and LM-CLR analysis. Each dot represents a significant ASV-SNP pair in the analysis using the default pseudo-count. **c**, Relative abundances of ASV2 stratified by groups of individuals based on the rs7780824 genotype. Data points are represented as dots. The boxplots show the median (center line), the first and third quartiles (box edges), and whiskers extending to 1.5 times the interquartile range. Violin plots illustrate the density distribution of the data points.

Supplementary Information For:

PALM: A quasi-Poisson regression approach for large-scale microbiome association studies and meta-analyses

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Supplementary Text

1.1 Justification of $E(W_{ik}) \approx E(W_{ik} \mid O_i)$ in PALM framework

Using the law of total expectation and properties of conditional expectation, we can express the difference between $E(W_{ik} \mid O_i)$ and $E(W_{ik})$ as

$$\delta_{ik} = E(W_{ik} \mid O_i) - E(W_{ik}) = \frac{\text{Cov}(W_{ik}, O_i)}{\text{Var}(O_i)} (O_i - E(O_i)).$$

Based on the relationship $O_i = \sum_{k=1}^K W_{ik}$, we can express $\text{Cov}(W_{ik}, O_i)$ and $\text{Var}(O_i)$ as

$$\text{Cov}(W_{ik}, O_i) = \text{Var}(W_{ik}) + \sum_{j \neq k} \text{Cov}(W_{ik}, W_{ij}),$$

$$\text{Var}(O_i) = \sum_{k=1}^K \text{Var}(W_{ik}) + 2 \sum_{1 \leq j < k \leq K} \text{Cov}(W_{ik}, W_{ij}).$$

We now establish conditions for $\delta_{ik} \rightarrow 0$ when $K \rightarrow \infty$:

1. Weak dependence between W_k and W_j for $j \neq k$. Under this condition, the cross-covariance terms are small; therefore $\text{Cov}(W_{ik}, O_i) \approx \text{Var}(W_{ik})$
2. W_{ik} is a small component of O_i . Under this condition, the variability of W_{ik} has small impact on the total O_i . Specifically, $\frac{\text{Var}(W_{ik})}{\text{Var}(O_i)} \rightarrow 0$ as $K \rightarrow \infty$.

These conditions cover a broad range of realistic scenarios and are satisfied in typical microbiome data.

1.2 Derivation of the variance of $\widehat{\beta}_k$ in PALM framework

Let V_k and V_k^* denote the variance of $\widehat{\beta}_k$ and $\widehat{\beta}_k^*$, respectively. Since $\widehat{\beta}_k = \widehat{\beta}_k^* - \text{median}\{\widehat{\beta}_k^*\}_k$, we have $V_k \approx V_k^* + \text{Var}(\text{median}\{\widehat{\beta}_k^*\}_k)$, provided that the dependence among $\widehat{\beta}_k^*$ is weak. We derive the formula for $\text{Var}(\text{median}\{\widehat{\beta}_k^*\}_k)$ as follows.

From quasi-likelihood theory¹, the estimate $\widehat{\beta}_k^*$ approximately follows a normal distribution with mean $\beta_k^* = \beta_k - \beta_O$ and variance estimated by V_k^* . For large K , the sample median is asymptotically normally distributed²

$$\text{median}\{\widehat{\beta}_k^*\}_k \sim \mathcal{N}(\beta_{\text{median}}^*, \text{Var}(\text{median}\{\widehat{\beta}_k^*\}_k)),$$

where $\beta_{\text{median}}^* = \text{median}\{\beta_k^*\}_k$. For normally distributed $\widehat{\beta}_k^*$'s, the $\text{Var}(\text{median}\{\widehat{\beta}_k^*\}_k)$ can be approximated as

$$\text{Var}(\text{median}\{\widehat{\beta}_k^*\}_k) \approx \frac{1}{4K \left(\frac{1}{K} \sum_{k=1}^K f_k(\beta_{\text{median}}^*) \right)^2},$$

where $f_k(\beta_{\text{median}}^*)$ is the normal density function of $\widehat{\beta}_k^*$ evaluated at β_{median}^* . By substituting the density function and simplifying the formula, we have

$$\text{Var}(\text{median}\{\widehat{\beta}_k^*\}_k) \approx \frac{K}{4 \left(\sum_{k=1}^K \frac{1}{\sqrt{2\pi V_k^*}} \exp\left(-\frac{(\beta_{\text{median}}^* - \beta_k^*)^2}{2V_k^*}\right) \right)^2}.$$

Under the sparse signal assumption that the majority of the features have no AA-level effects (i.e., $\beta_k = 0$ for most k), the majority of β_k^* shares the same mean $-\beta_O$, leading to $\beta_{\text{median}}^* \approx -\beta_O$. Therefore, the exponential term approaches 1 for most k , resulting in the final variance approximation

$$\text{Var}(\text{median}\{\widehat{\beta}_k^*\}_k) \approx \frac{K}{4 \left(\sum_{k=1}^K \frac{1}{\sqrt{2\pi V_k^*}} \right)^2}.$$

1.3 Generating PALM summary statistics for a study with correlated samples

Suppose we have n clusters in the data. Let i index clusters ($i = 1, \dots, n$) and t index correlated samples within a particular cluster ($t = 1 \dots, T_i$). The notation for data is defined similarly to the independent sample case, but subscript i is replaced with it to index sample t within cluster i . The quasi-score function can be written as

$$\mathbf{S}(\boldsymbol{\theta}_k) = \sum_{i=1}^n \sum_{t=1}^{T_i} \left(Y_{itk} - \mu_{itk} + \frac{1}{2} h_{itk} \right) \mathbf{X}_{it}.$$

Let $(S_1, \mathbf{S}_0)$ be the partition of $\mathbf{S}(\boldsymbol{\theta}_k)$ corresponding to the parameter of interest β_k^* and the nuisance parameters. We estimate $\tilde{\boldsymbol{\theta}}_k$, construct $\hat{\beta}_k^*$, and calculate $\begin{bmatrix} \mathcal{I}_{11} & \mathcal{I}_{10} \\ \mathcal{I}_{01} & \mathcal{I}_{11} \end{bmatrix}$ and $\mathcal{I}_1$ using the same procedures described above. The sandwich estimator for the variance of $\hat{\beta}_k^*$ is adjusted to account for correlated samples as

$$V_k^* = \mathcal{I}_1^{-1}(\tilde{\boldsymbol{\theta}}_k) \left(\sum_{i=1}^n \left(\sum_{t=1}^{T_i} U_{it}(\tilde{\boldsymbol{\theta}}_k) \right) \left(\sum_{t=1}^{T_i} U_{it}(\tilde{\boldsymbol{\theta}}_k) \right)^\top \right) \mathcal{I}_1^{-1}(\tilde{\boldsymbol{\theta}}_k),$$

where $U_{it} = S_{1,it} - \mathcal{I}_{10} \mathcal{I}_{00}^{-1} \mathbf{S}_{0,it}$, $S_{1,it}$ and $\mathbf{S}_{0,it}$ are the it th summand of S_1 and $\mathbf{S}_0$, respectively.

1.4 Comparison of methods

We compared the performance of PALM with ANCOM-BC2, LinDA, LM-CLR, and MaAsLin2. In all meta-analyses, both in simulations and real data applications, summary statistics for each method were generated at the individual study level. A fixed-effect meta-analysis was then performed by combining these summary statistics for association testing, and microbial features associated with the covariate of interest were identified using the BH procedure with a target FDR of 0.05. All analyses were conducted in R (v4.3.1). Feature prevalence filters were applied externally, and no additional prevalence filters were applied within each method's function.

- ANCOM-BC2: we used the `ancombc` function in the package ANCOMBC (v2.2.2) with the specified argument `prv_cut = 0`. Summary statistics `lfc_labels1` and `se_labels1` were extracted from the output.
- LinDA: we used the `linda` function in the package MicrobiomeStat (v1.2) with the specified arguments `feature.dat.type = 'count'`. Summary statistics `log2FoldChange` and `lfcSE` were extracted from the output.
- LM-CLR: we first added a pseudo-count (default: 0.5) to each cell of the count matrix and then applied the centered log-ratio transformation. To analyze a study with independent samples, we used the `lm` function available through the base R distribution for each transformed feature variable as the response. To analyze a study with correlated samples, we used the `lmer` function in the package lme4 (v1.1-34). Summary statistics were then extracted from the coefficient table of the output.
- MaAsLin2: we used the `Maaslin2` function in the package Maaslin2 (v1.14.1) with the specified arguments `min_prevalence = 0`, `transform = 'AST'`. Summary statistics `coef` and `stderr` were extracted from the output.

For all methods, the fixed-effect meta-analysis was performed using the function `rma` in the package metafor (v4.2-0) with specified argument `method='EE'`. The Cochran's Q heterogeneity test p -value was also extracted from the output.

1.5 Analyzed datasets

Colorectal cancer (CRC) studies: The five studies include subjects from five different countries. The sample sizes of the studies are 109, 127, 120, 114, and 104 (*SI Appendix*, Table S1). All raw sequencing data across studies were reprocessed using the same bioinformatics pipeline for taxonomic profiling³. We removed two samples with an outlier sequencing depth of less than 2,000. Our analysis focused on $K = 401$ species after removing those with a prevalence of less than 20% (prevalence was defined as the percentage of non-zero counts in all samples across studies).

Metabolome (MTBL) studies: In all studies, the microbiome and metabolome were profiled from the same fecal samples. The datasets were curated and processed as part of an established effort⁴. We used the eight adult studies from this resource, with sample sizes of 96, 294, 240, 220, 444, 382, 287, and 164. These studies exhibit notable heterogeneity in study design, cohort demographics, and technical aspects of microbiome and metabolite measurements (*SI Appendix*, Table S3). Our analysis focused on 450 HMDB-annotated metabolites with study-specific prevalence $\geq 10\%$ in at least two studies and 101 genera with study-specific prevalence $\geq 10\%$ in at least one study. Following standard practice for normalizing skewed metabolite data, we applied a rank-based inverse normal transformation to the metabolite variables.

Host genetics study: The study is based on a population-based birth cohort of term-born and healthy infants⁵. The host genetics and microbiome data were collected from a subset of the cohort with 502 infants aged between 2 and 242 days. Most of the infants were non-Hispanic White. Genotypes were determined using the Illumina Genome Diversity Array (GDA, 8v1.0) at the Center for Inherited Disease Research (CIDR) (Baltimore, MD). Individuals were removed if they met any of the following quality control (QC) criteria: genotype vs. self-reported sex mismatches, unexpected duplicate samples, $> 10\%$ missing genotypes, and autosomal heterozygosity exceeding five standard deviations ($SD=0.8\%$) above or below the mean of 30.4% . SNPs meeting any of the following QC were excluded: $> 10\%$ missing genotypes, exact Hardy-Weinberg p -value $< 5 \times 10^{-8}$, or genotype concordance $< 90\%$. A total of 1,785,122 directly genotyped SNPs remained following QC. Ancestry principal components were estimated from autosomal SNPs with $MAF > 0.005$ and $LD\ r^2 < 0.10$, using the 1,000 Genomes dataset as the reference. Imputation was performed using the TopMed reference panel (r^2 version 1.0.0) of $>100,000$ whole genome sequences (human genome build 38) and the TopMed Imputation Server (version 1.7.3). Specifically, Eagle (version 2.4) was used for genotype phasing and minimac4 (version

1.0.2) was used for genotype imputation. Genotype dosages for 6,174,920 biallelic SNPs with MAF > 5% and imputation INFO score > 0.9 were included in the analysis.

The details of microbiome sample collection, processing, and sequencing were previously reported⁶ and were briefly summarized here. For the characterization of the gut microbiome, we extracted bacterial DNA from the stool samples collected at enrollment using the DNeasy PowerSoil Kit (Qiagen). The libraries were then sequenced on an Illumina MiSeq with 2×250 base pair reads. The 16S rRNA sequences were processed using the dada2 package, grouping the sequences into ASVs and assigning taxonomy based on the SILVA reference database. Our analysis focused on 109 ASVs with a prevalence greater than 5%.

2. Supplementary Figures and Tables

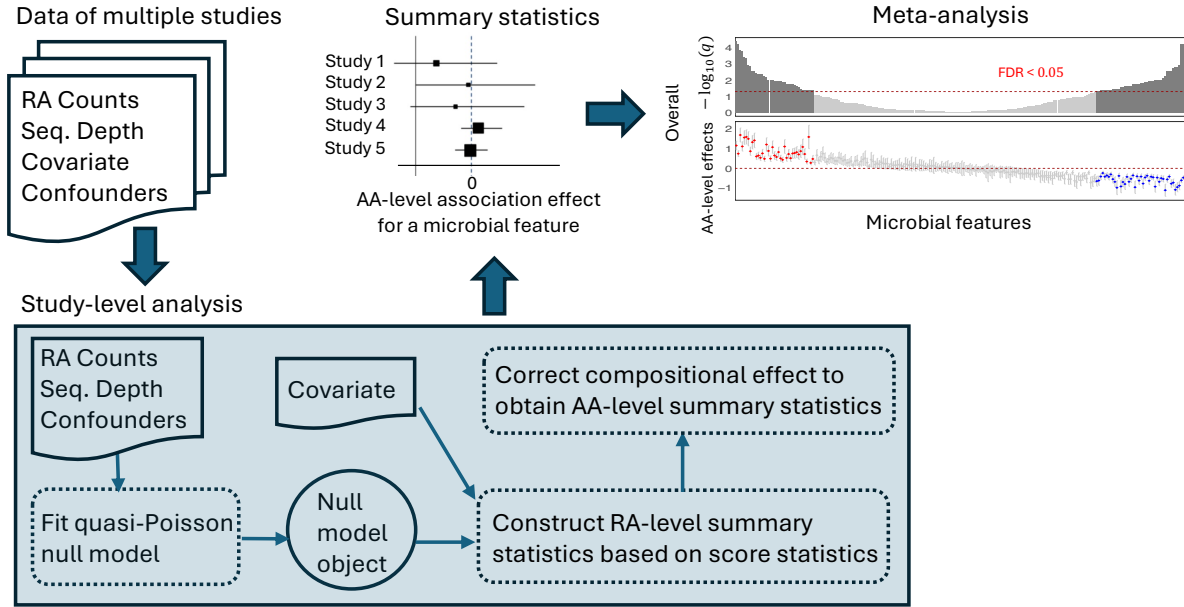


Fig. S1: Schematic overview of PALM for meta-analysis. For each study, PALM takes RA count data of microbial features, sequencing depth, the covariate of interest, and any confounders that need to be adjusted for as input to generate AA-level summary statistics. Specifically, for each feature, PALM first fits a quasi-Poisson null model (without the covariate), then calculates score statistics based on the null model output and the covariate, and finally constructs the RA-level summary statistics (i.e., RA association effect estimate and variance) based on the score statistics. Next, PALM corrects the compositional effect using the RA-level summary statistics of all features to derive AA-level summary statistics. The forest plot illustrates the AA-level summary statistics for a feature. These summary statistics can be combined via meta-analysis to obtain the overall effect estimate, variance, and p -value. Covariate-associated microbial features are selected based on the test p -values, accounting for multiple testing across features.

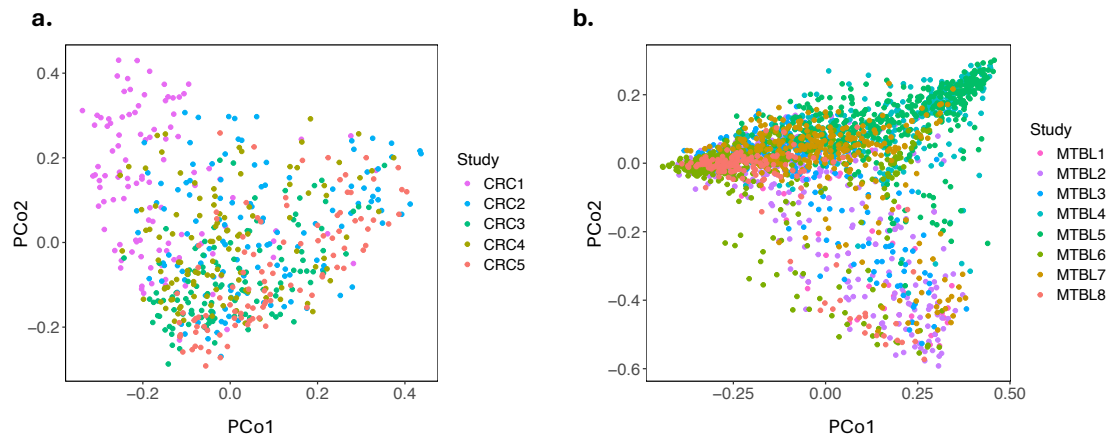


Fig. S2: Principal coordinate plots of real microbiome data over studies in the two meta-analysis applications. **a**, Meta-analysis of five microbiome-colorectal cancer (CRC) association studies. **b**, Meta-analysis of eight microbiome-metabolome (MTBL) association studies. Bray-Curtis dissimilarity was used in the principal coordinate analysis. Data points from different studies are color-coded. The PERMANOVA test p -values for testing the study effect in panels a and b are less than 10^{-5} .

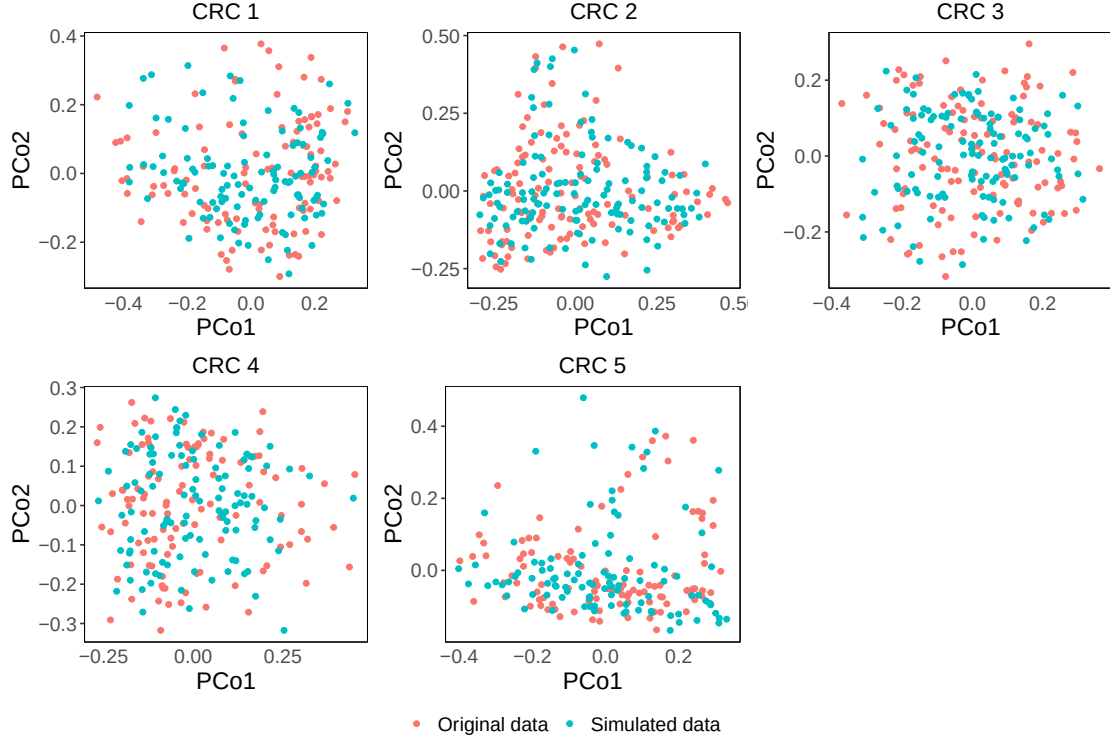


Fig. S3: Principal coordinate plots for the original and simulated microbiome data. Each panel represents a single study. The original data refers to the real data in a colorectal cancer (CRC) study (indicated in the panel title), and the simulated data was generated to mimic the original data using the MIDASim simulator. Bray-Curtis dissimilarity was used in the principal coordinate analysis. The PERMANOVA test p -values for testing the difference between the original and simulated data exceed 0.5, indicating a close resemblance between the two datasets.

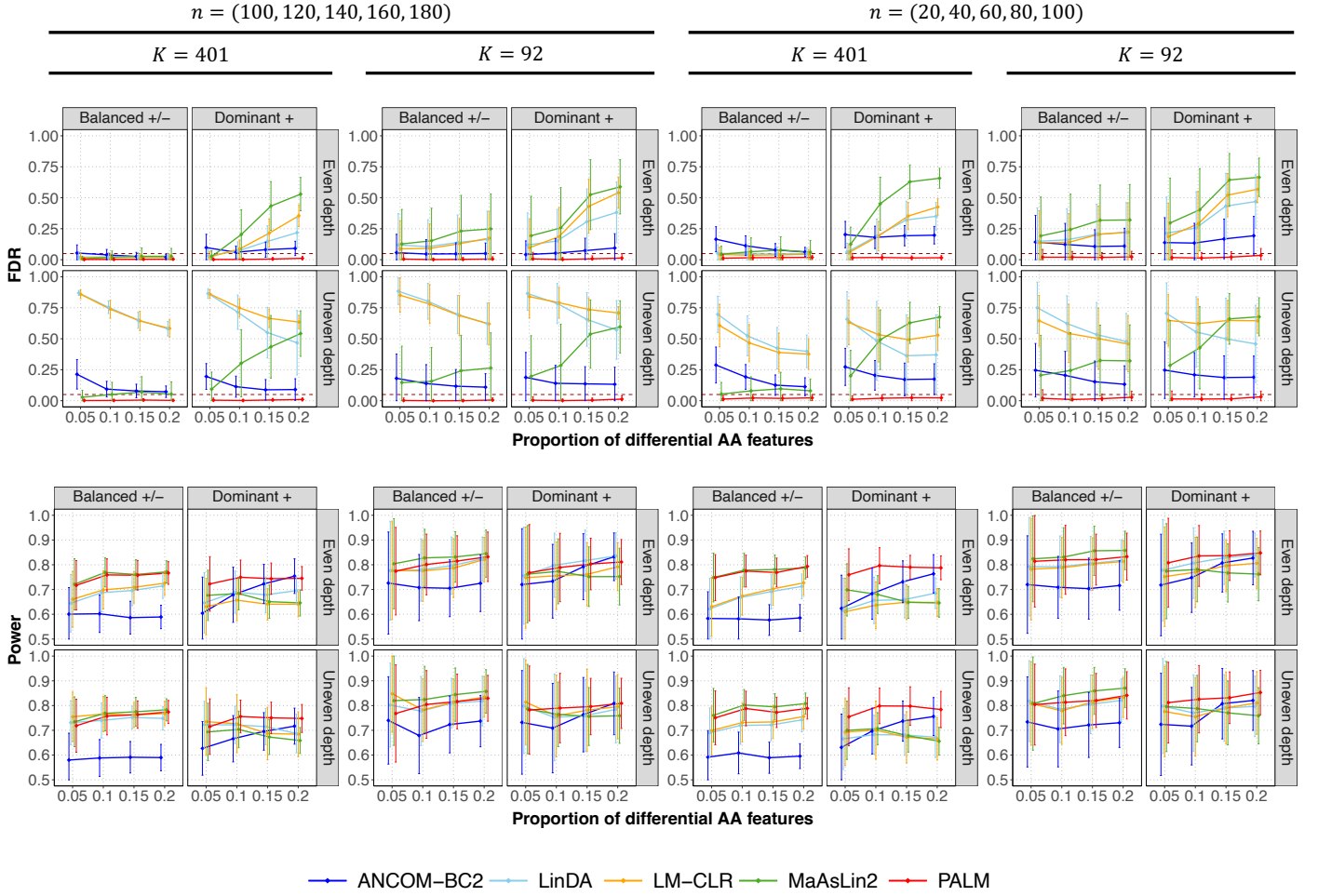


Fig. S4: Comparison of PALM meta-analysis with pooled-data analysis by other methods across five simulated studies with independent samples. The two sections of the figure show empirical FDR (top, the horizontal line indicates the target FDR) and power (bottom). The x-axis represents the proportion of differential absolute abundance (AA) features. The top title shows the two scenarios of sample sizes n in five studies. The second title shows the two scenarios of feature size K . The column facet title represents two effect direction scenarios: Balanced +/-, where differential AA features have an equal probability of exhibiting positive or negative effects, and Dominant +, where all differential AA features exhibit positive effects. The row facet title represents two sequencing depth scenarios: one with even sequencing depth and one with uneven sequencing depth between the comparison groups. Each panel displays the mean estimated metrics with $\pm$ standard errors (indicated by error bars) based on 100 simulation replicates.

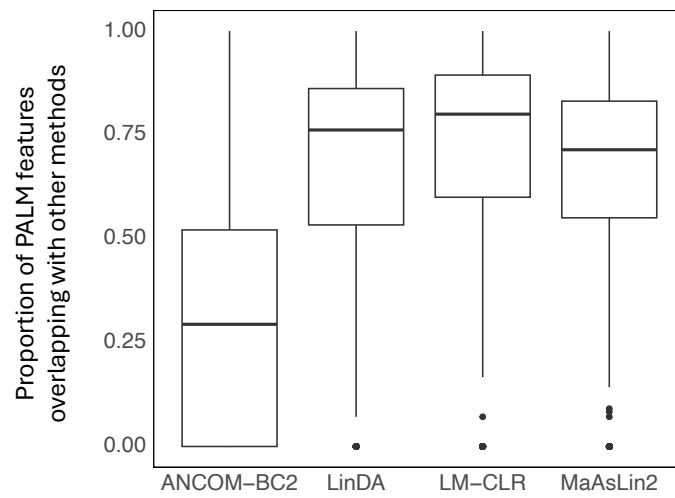
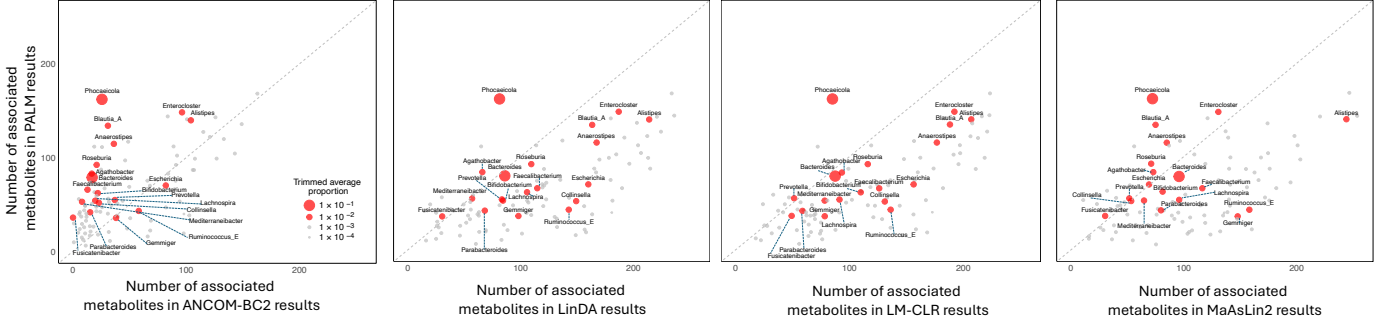
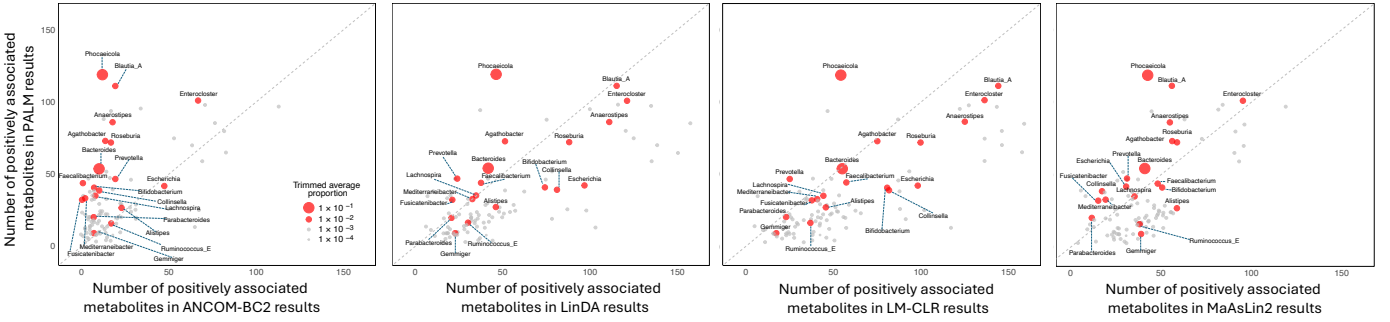


Fig. S5: Overlap between microbial features identified by PALM and those identified by other methods in the meta-analysis of microbiome-metabolome association studies. The x-axis lists the individual methods being compared. The y-axis shows the proportion of PALM-identified metabolite-associated features that overlap with those identified by other methods across all metabolites. Each boxplot indicates the median (center line), the first and third quartiles (box edges), whiskers extending to 1.5 times the interquartile range, and individual outliers (dots).

a. All Associations



b. Positive Associations



c. Negative Associations

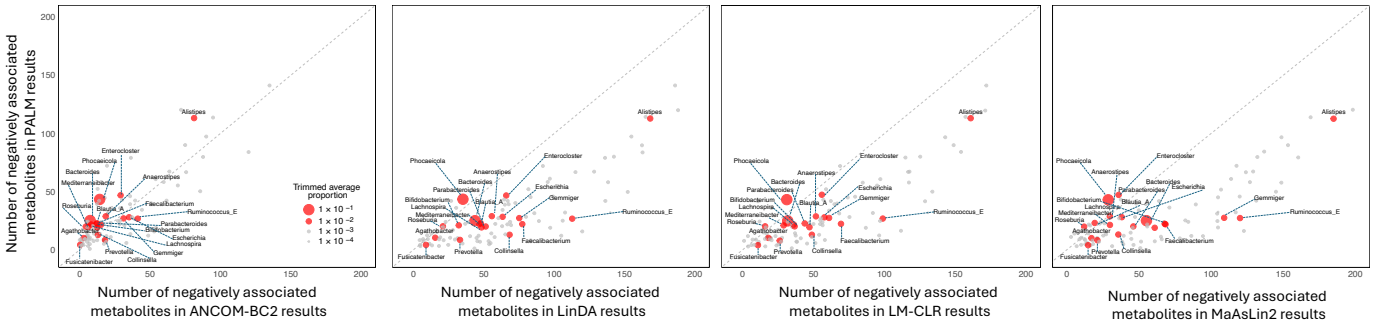


Fig. S6: Number of associated metabolites per microbial feature in PALM results compared to other methods' results from the meta-analysis of microbiome-metabolome association studies. a, number of all associations. b, number of positive associations. c, number of negative associations. Each dot represents a feature, with dot size proportional to the trimmed average proportion of the feature across all individuals in all studies. Features with an average proportion ≥ 0.01 are shown in red and annotated with their feature names, while other features are shown in gray. Compared to other methods, the top features in PALM results tend to have higher average proportions across all individuals in all studies (i.e., in **a**, the features that are well below diagonal lines have low proportions, while features closer to or above the diagonal line have higher proportions). This trend is driven by positive associations, rather than negative associations (see **b**, **c**).

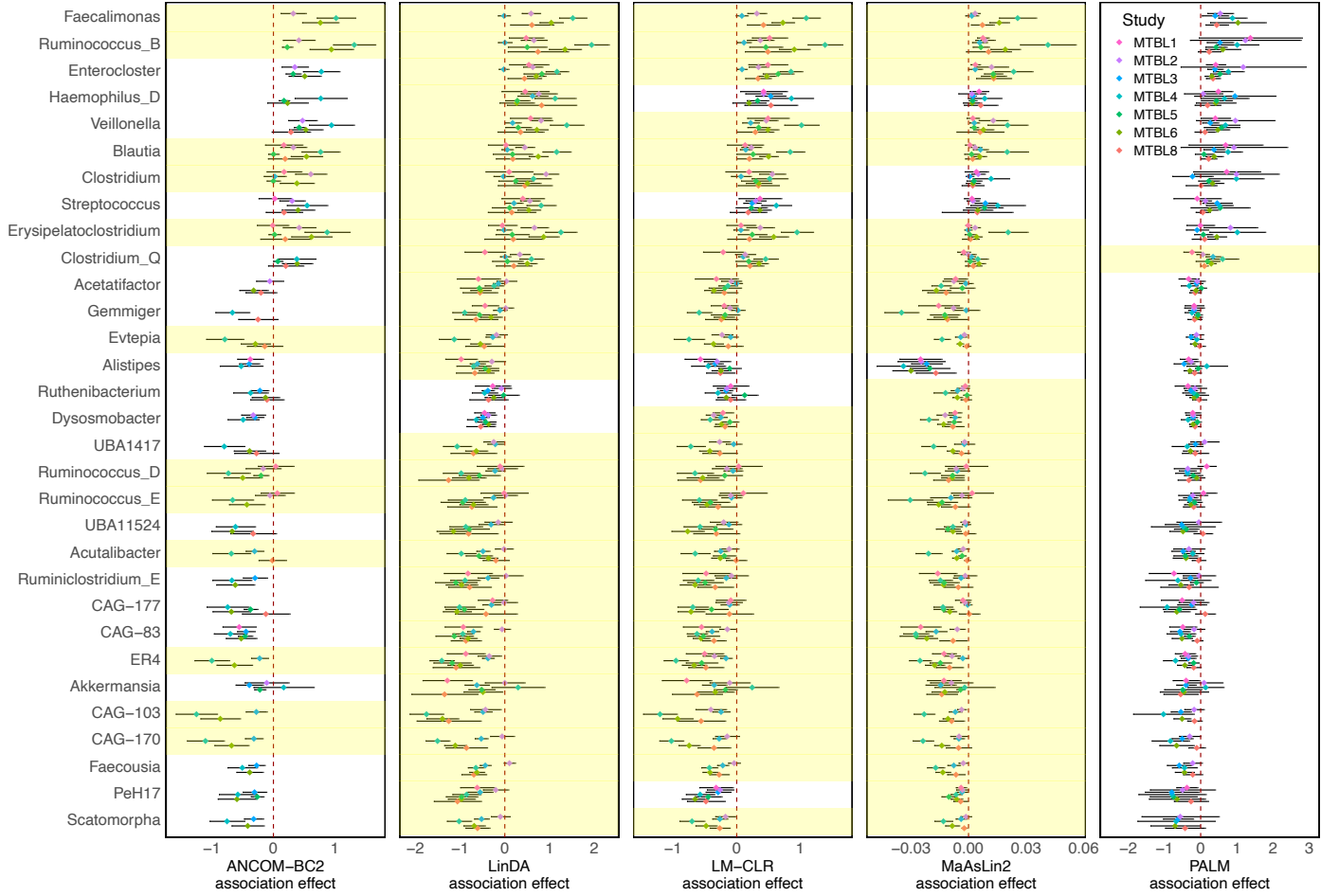


Fig. S7: Forest plots for a set of microbial features associated with the metabolite cholic acid in the meta-analysis of microbiome-metabolome (MTBL) association studies. x-axis represents a core set of cholic-acid-associated features consistently identified by all methods but exhibiting significant heterogeneity in effects across studies in at least one method. Each panel shows the summary statistics of a method, including the association effect estimates across studies (color-coded dots) and their 95% confidence intervals (black lines). The row for features with significant heterogeneous effects across studies is shaded yellow in each panel (Cochran's Q test q -value < 0.1);

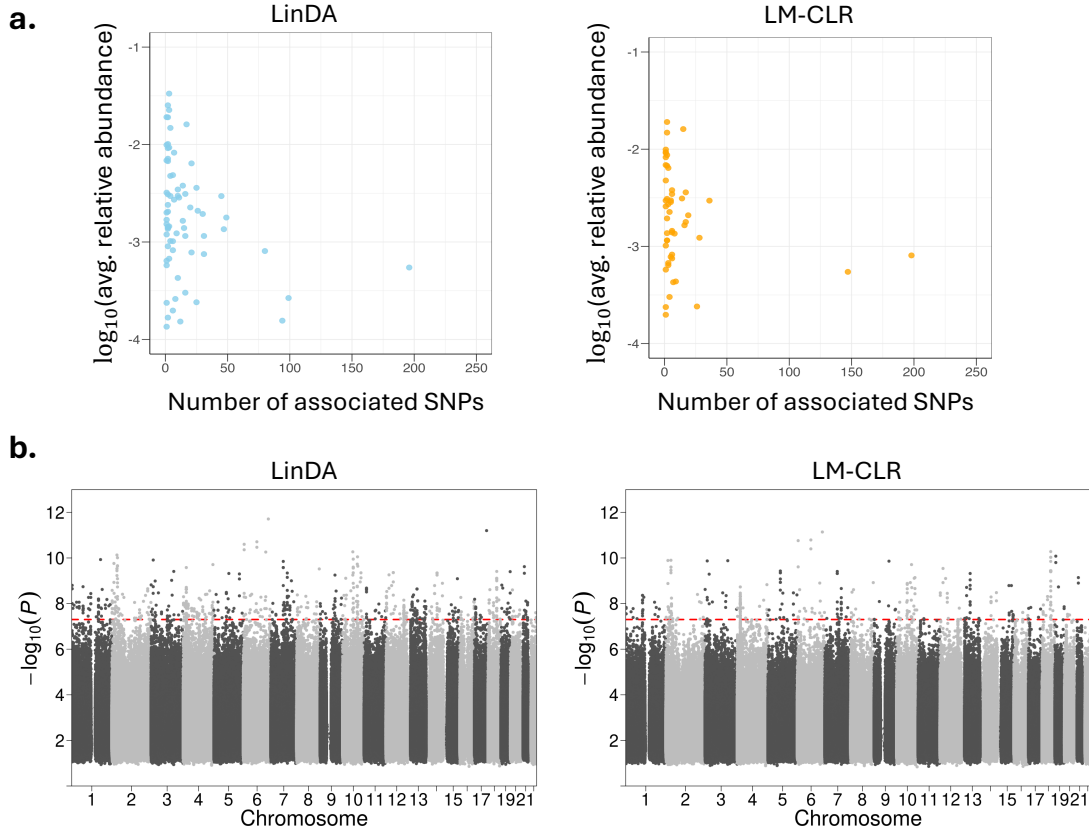


Fig. S8: Additional microbiome and host genetics GWAS (mbGWAS) results from LinDA and LM-CLR. **a**, Summary of the LinDA and LM-CLR identified SNP-associated ASVs (dots) shown in Fig. 6. The x-axis shows the number of SNPs associated with each ASV, and the y-axis shows the $-\log_{10}$ average relative abundance of each ASV. **b**, mbGWAS Manhattan plots for LinDA and LM-CLR when the pseudo-count of 0.01 was used in the analysis. Each SNP was tested against each of the 109 ASVs and the Manhattan plot shows the lowest resulting p -value for each SNP. The horizontal dashed line marks the genome-wide significance ($p\text{-value} = 5 \times 10^{-8}$).

Table S1: Datasets of microbiome association studies for colorectal cancer.

Study ID	Country	No. Case	No. Control
CRC1 ⁷	Austria	46	63
CRC2 ⁸	China	73	54
CRC3 ³	Germany	60	60
CRC4 ⁹	France	53	61
CRC5 ¹⁰	United States	52	52

Table S3: Datasets of microbiome-metabolome association studies.

Study ID	Correlated Sample Y/N	No. Subjects Control/Case	No. Samples Control/Case	Microbiome 16S/WGS	Metabolome Target/Untargeted
MTBL1 ¹¹	N	54/42	54/42	WGS	Targeted
MTBL2 ¹²	N	74*/220	74*/220	WGS	Targeted
MTBL3 ¹³	N	102/138	102/138	16S	Untargeted
MTBL4 ¹⁴	N	56/164	56/164	WGS	Untargeted
MTBL5 ¹⁵	Y	24/51	139/305	WGS	Targeted
MTBL6 ¹⁶	Y	26/79	104/278	WGS	Untargeted
MTBL7 ¹⁷	N	67/220	67/220	WGS	Untargeted
MTBL8 ¹⁸	Y	83/NA	164/NA	16S	Untargeted

*: the number is after removing control subjects shared with MTBL1.

Tables S2 and S4 are too big and included in a separate Excel file.

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